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Laying the foundations for context-aware and AI-ready fault diagnosis with the Operations Ontology

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Abstract.
In wind farm operations, the full value of operational data is not realised due to obstacles in understanding and integration. With vast amounts of data in operational silos, reference ontologies provide a common framework for describing and connecting heterogeneous data, establishing a shared meaning that is machine-readable, human-understandable and a foundation for applying AI. As part of IEA Wind Task 43 and within the WeDoWind ecosystem, we have formed a public working group of experts across multiple disciplines to develop a foundational ontology of operations. Starting with foundational entities in the field of operations and maintenance, such as *maintenance process* and *alarm system*, we develop a section of the Operations Ontology (OpOn), focusing on a demonstration use case involving diagnostics and troubleshooting of rotor over-speed protection alarms. Here we illustrate how data annotation and integration become more intuitive and efficient with the use of the ontology, and we describe how this builds a solid foundation for the use of modern AI.

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1 Introduction

Throughout the operational phase of a wind farm, vast amounts of diverse data are processed by many different stakeholders. Supervisory Control and Data Acquisition (SCADA) data, Condition Monitoring System (CMS) data, work orders, service and inspection reports, technical specifications, maintenance manuals, weather forecasts and more contribute to a rich, but fragmented, information landscape. Improving the clarity, consistency and interoperability of data structures and semantics allows stakeholders to extract greater actionable value from their data. The ETIPWind Strategic Research and Innovation Agenda highlights the optimisation and further digitalisation of operations and maintenance (O&M) as one of its core research and innovation priorities [1]. The challenges it identifies – standardised data exchange across wind farm subsystems, AI-driven predictive maintenance, and digital ecosystems connecting sensor data to lifecycle decision-making – represent important opportunities for the wind industry. Aligned with these priorities, we identify the following areas where significant value remains to be realised:

- Integration of operational data streams – including SCADA alarms, other events and metric signals – into unified diagnostic frameworks that support systematic fault diagnosis and troubleshooting;
- Semantic linking of SCADA event, alarm and metric signal taxonomies to the technical information about the components and systems to which they relate, providing machine-readable semantics and context to drive intelligent use of the observational data;
- Embedding CMS datasets into day-to-day operational monitoring and site operation workflows, closing the gap between diagnostic/predictive capability and operational practice; and
- Structured linking of work orders, service reports and inspection records to alarm and outage logs, creating a holistic view to facilitate automated benchmarking, reporting and optimisation.

Several existing standards are available to support data interoperability in the wind turbine O&M space. For example, the ISO/IEC 81346 Reference Designation System (RDS) standards, together with the TIM Wind guidelines, provide a framework for structured system identification and breakdown, and the IEC 61850/61400-25 series addresses communication protocols and information models for wind power plants. However, these standards were designed for specific purposes – structured naming and communication interoperability, respectively – rather than as general frameworks for semantic integration across the full range of operational data types. In practice, data from different sources, vendors and systems remains difficult to combine, even when relevant standards are applied, because the underlying meaning of the data is not captured in a form that is both explicit and machine-processable.

In any functioning information system, the meaning of the data is captured somewhere – whether implicitly in application logic and user knowledge, or explicitly in data descriptions and annotations. When datasets are stored without semantic description, the knowledge required to interpret them remains within those systems, or is tacit knowledge held by the people who operate those systems. When such datasets are shared with an external party or system, significant knowledge is typically lost. The approach of using ontologies to describe data can be understood as an effort to move from implicit semantics, to explicit semantics encoded in the data itself.

The use of ontologies is a proven approach to mitigating data interoperability challenges, with a particularly successful track record in the biological and biomedical domains [2]. Whilst information systems with their software and databases vary significantly across implementations and evolve quickly, ontologies provide shared and stable representations of domains. When developers use common ontologies for data annotation, different systems can more easily exchange and aggregate data. Beyond data integration, ontologies capture domain knowledge in a form that enables automated logical reasoning and provides a basis for symbolic AI [3, 4]. This is particularly relevant in the context of emerging hybrid AI approaches that combine symbolic reasoning with statistical models such as large language models [5], where ontologies can supply the structured domain knowledge that data-driven general-purpose models lack.

1.1 Gap and motivation

A recent review of knowledge engineering for the wind energy industry found that no comprehensive industry-specific reference ontologies are available [6]. Despite earlier research in the field of wind energy semantic artefacts identifying the potential of ontologies to process natural language and support knowledge-based systems [7] and to support automated question-answering systems [8, 9], the adoption of such artefacts outside of academia has been limited. A suite of interoperable reference ontologies for wind farm operations could provide a stable, shared vocabulary for stakeholders, which complements existing standards by providing the broader semantic layer that current standards do not address.

1.2 Contribution
To address this gap, the Operations Ontology group was formed within IEA Wind Task 43[1] and the WeDoWind Information Modelling Community[2], bringing together practitioners and researchers from wind farm operators, consultancies, software providers, research institutions and universities[3] to collaboratively develop ontologies that support the operational phase. The Operations Ontology (OpOn) aims to provide a foundational semantic framework for integrating the diverse data types encountered in wind energy operations. The objectives of our effort are organised around three complementary goals.

1. For data integration and interoperability:
   - Provide a clear and consistent framework for semantically describing all O&M-relevant data;
   - Enable better preservation of semantics when datasets are shared between parties and systems;
   - Support automated data processing workflows.
2. For usability and adoption:
   - Facilitate intuitive and efficient data annotation (semantic enrichment);
   - Allow non-experts to more easily understand and navigate data relevant to O&M;
   - Enable tools for automated checking of semantic consistency; and
   - Shift semantic encoding from software logic to data, making both more easily interpretable.
3. For AI-readiness:
   - Support advanced automated ontology-based reasoning; and
   - Support the provision of appropriate context to Large Language Models (LLMs) and other relevant connectionist models via knowledge grounding, use of Retrieval Augmented Generation (RAG), Model Context Protocol (MCP), or other techniques.

In this paper, we present an overview of the current state of the Operations Ontology, demonstrating its practical application through a use case involving diagnosis and troubleshooting of rotor over-speed protection alarms. We illustrate how the ontology supports structured, context-aware fault diagnosis and provides a foundation for AI-assisted decision support.

The remainder of the paper is structured as follows. Section 2 explains how we created a practice-near but consistent ontology slice in a relatively short amount of time. Section 3 presents the context and a technical overview of the troubleshooting use case, setting the scope and providing examples for the introduction of the Operations Ontology in Section 4. We return to the use case in Section 5 to apply ontology-based diagnostics, illustrating how the ontology can be used to provide the necessary context to resolve the issues. In Section 6, we discuss the role of AI. In Finally, Section 7 concludes with a summary and discussion of future work.

2 Methods
The domain was scoped and mid-level classes from the O&M domain were identified based on the practical requirements of stakeholders. These classes were mapped upwards to align with a top-level ontology and iteratively refined downwards to achieve an appropriate level of granularity, whilst leaving scope for organisations to build specific extensions that reflect their contexts.

Drawing on recommended practices developed by IEA Wind Task 43 for ontology creation, publication, and maintenance for wind energy domain experts [10], the OpOn aligns with existing relevant ontologies. Basic Formal Ontology (BFO) provides the top level, domain-neutral framework for refining fundamental categories [2]. Reusing only relevant classes from appropriate mid-level, BFO-aligned ontologies, such as Common Core Ontologies (CCO)[4], promotes interoperability and minimises ontological overcommitment, which is the unnecessary complexity that arises when more entities are included in an ontology than are necessary to explain the domain.

[1] https://iea-wind.org/task43

[2] https://community.wedowind.ch/spaces/20886575

[3] We aim to further broaden the degree of representation from across the industry in the future, for example to also include wind turbine OEMs.

[4] https://github.com/CommonCoreOntology/CommonCoreOntologies

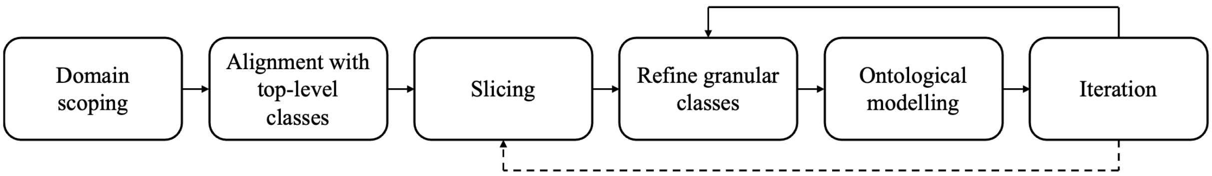

Figure 1: Ontology development process

The scope of the ontology was defined based on the following requirements and constraints:

- It should provide a clear and consistent framework for describing all types of operational data. The ontology should not cover specific details of each type, but should be general enough to accommodate them. If this requirement is not met, different frameworks would be needed for different types of data, hindering interoperability.
- It should focus on uncontroversial and stable content. Terms that have contextually-specific definitions are best captured at application-level.
- It should be as small as possible, whilst meeting core objectives. Meeting this requirement facilitates a short and efficient development cycle, as well as improving ease of adoption and maintainability.
- It should be clear, intuitive and easy to understand. Meeting this criterion is critical for community adoption beyond the working group.

Competency questions (CQs) are questions in natural language that reflect the requirements and scope of an ontology. In the development of the Operations Ontology, the primary focus was developing CQs that reflect tasks for stakeholders working within wind turbine O&M. Illustrative application-focussed competency questions for this ontology are presented in Table 1.

Competency Questions
In a situation where a turbine has stopped with a rotor over-speed protection alarm, what is the best course of action based on all available technical documentation and all relevant historical experience across my fleet?
Considering recent issues with downtime due to false positive rotor over-speed protection alarms, what corrective and preventative actions should I take across my fleet?

Table 1: Example Competency Questions

The OpOn was developed in Protégé and is expressed in Web Ontology Language (OWL), the most common standard for knowledge representation in semantic technologies. It has been designed to be interoperable with emerging domain ontologies in the wind field, such as Wind Energy Ontology (WEO), also emerging from IEA Wind Task 43. In this early phase of development, the ontology is evaluated based on its ability to answer the competency questions and through consensus among the team of ontology developers. We acknowledge that future development requires metrics to assess when the ontology can be considered stable, which requires a mixture of internal evaluation metrics and external validation.

3 Use case overview: diagnosis and troubleshooting of over-speed protection alarms

We apply the OpOn to an example case of a turbine stopped with one or more rotor over-speed protection alarms to illustrate the benefit of an Operations Ontology. Whilst this is a small-scale example, the practical benefit of ontology-based solutions lies at scale when large volumes of heterogeneous data spanning different turbine platforms, teams, business units and organisations are connected to address a wide range of scenarios. The ontology is used to make tacit organisational knowledge explicit, allowing teams to cumulatively add information to the same context. This makes it easier to later build applications that couple data in a way that could not have been envisioned from the start.

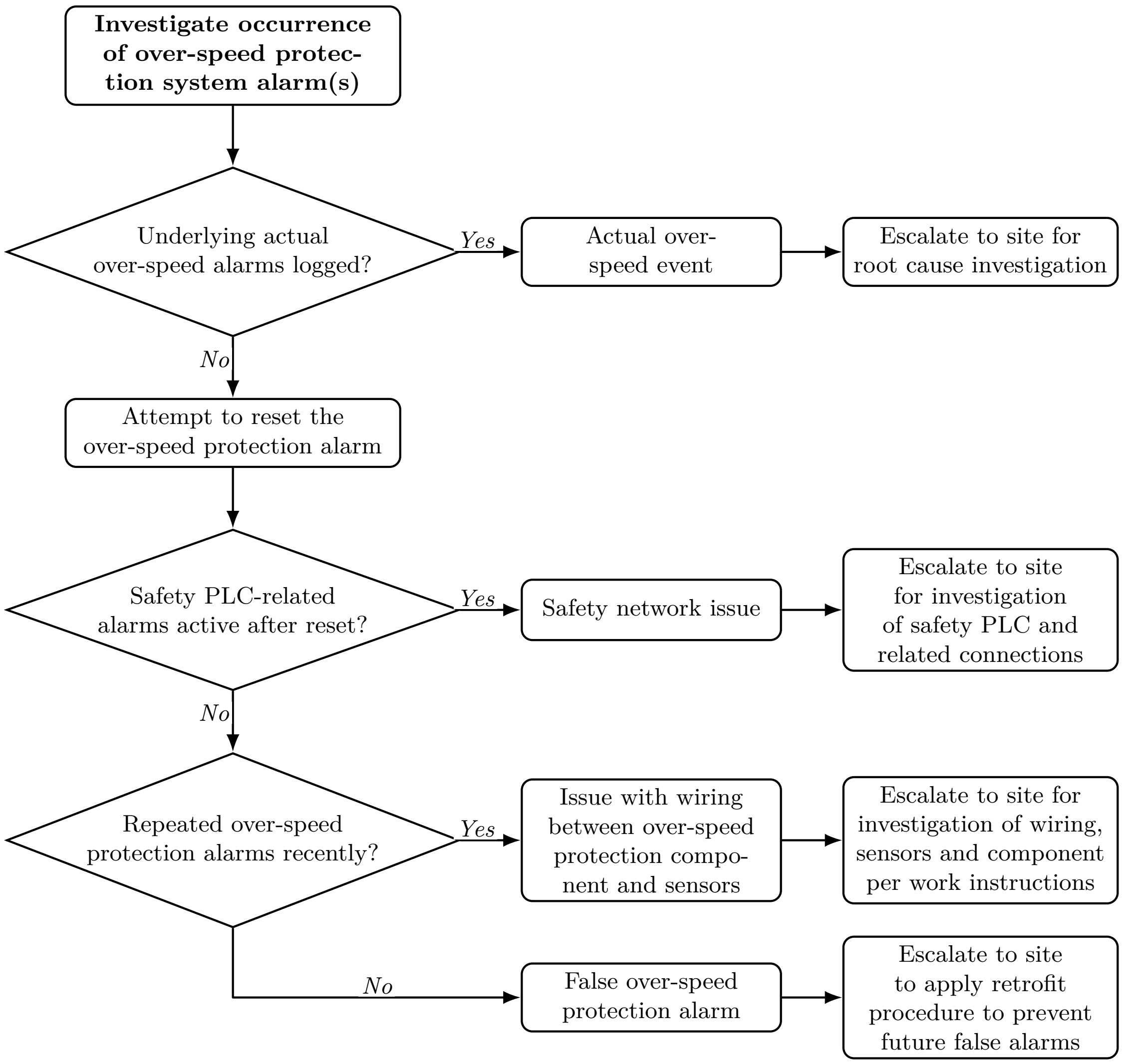

Figure 2: Rotor over-speed protection system alarm troubleshooting process

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3.1 The rotor over-speed protection system and associated data
Rotor over-speed protection with redundant independent systems is a necessary safety feature in modern wind turbines [11] designed to prevent turbine over-speed for multiple reasons:

- Structural protection – prevent extreme loads, excessive blade tip speeds, excessive blade displacement risking a collision with the tower, excitation of tower resonant frequencies [12];
- Generator protection – prevent over-frequency/over-voltage operation which can lead to excessive thermal loading in particular for direct-drive generators; and
- Protection against loss of braking – for direct-drive turbines, a grid disconnect or converter fault removes generator counter-torque, leaving the rotor uncontrolled, and the rotor over-speed protection triggers pitch feathering and mechanical braking to prevent this.

The protection system itself is intricate and has multiple failure modes (for example, sensor failure, loose or degraded cabling, and network communication failure) that can be difficult to diagnose and require access to multiple data sources:

- SCADA data – to determine whether this is an actual over-speed event, the pitch positions, signal noise and turbine active power output;
- Event logs – recent and historical alarms and events from the turbine control system implicating related systems, such as the safety PLC, regular and/or safety communication networks;
- Work order history – previous similar and/or related issues, related maintenance activities and retrofits;
- Rotor over-speed protection system event log – detailed log of over-speed protection events, actions taken by the control system and when; and
- Remote operating procedures, work instructions, troubleshooting guides and other technical documentation.

3.2 Troubleshooting process
In the alarm troubleshooting process, the root cause needs to be determined from a range of possible causes, including an actual rotor over-speed event, a safety network communication issue, a false alarm from the over-speed protection system, etc., and the appropriate remedial action taken. In this use case, the troubleshooting process illustrated in Figure 2 should start whenever there are one or more active alarms that originate from the rotor over-speed protection system.

4 Ontology
This section introduces the Operations Ontology (OpOn), with focus on the entities involved in the use case introduced in Section 3. A full documentation of the current version is beyond the scope of this paper. In the absence of a suitable domain-neutral, BFO-aligned O&M ontology available for reuse, we have developed some such content.

4.1 Material entities
In BFO, material entities are concrete physical entities composed of matter. Within an O&M context, they include the operated pieces of equipment and systems with sub-components and sub-systems (e.g. wind farms, wind turbines, rotor systems, and rotor over-speed protection systems), equipment used in O&M processes (e.g. jack-up vessels for offshore maintenance), the external site environment (e.g. the air that drives the rotor through its kinetic energy), and the people involved in O&M.

BFO does not include the entity *system*, and different mid-level ontologies (e.g. IOF Core and CCO) have taken different approaches to dealing with systems. None of them, as far as we are aware, provide a sufficient representation for our purposes. We have defined a system as an object aggregate in which each object part fills a system role as a starting point for development.

We envisage a separate effort to develop a detailed wind turbine systems and components ontology, based upon existing relevant standards (e.g. ISO/IEC 81346). Such an ontology is relevant not only for O&M, but for the majority of the lifecycle stages. In the OpOn, we have preliminarily introduced systems and components required for the use case, and intend to move those into a separate ontology later.

4.2 Roles

In BFO, roles are externally grounded realisable entities, which specifically depend on their bearer. The perspective of roles helps us, for example, to represent stakeholders (e.g. *operator role*), contractual parties (e.g. *insurer role*) and the entities that are the subjects of contracts and agreements (e.g. *insured entity role*). During the operational phase, a wind turbine has an *operated entity role* in operational processes, *asset role* from the perspective of ownership and financial arrangements, and *product role* from the perspective of the purchase agreement and its warranty provisions.

Roles are important to representing system breakdowns, where we need to capture both the role of an individual entity (as in the previous examples mentioned) and the system role ("slot") that it fits into.

4.3 Processes

Processes are entities that unfold in time (i.e. occurrents) and have material entities as participants. The entire operational phase is a process centred around the realisation of the purpose for which an engineered artifact was designed and manufactured. It is composed of many different sub-types of *operational process*, including for example *troubleshooting process*. For the wind energy domain, we define more specific sub-types of these, such as *rotor over-speed troubleshooting process*.

Ontological relations are used to link the processes to the material entities that are their participants, to qualities and realisable entities involved, and to any information content produced or consumed. For example, in a troubleshooting process focused on rotor over-speed protection alarms, the safety system communication network participates in the capacity of being a subject of diagnosis and troubleshooting, and some human being (e.g. turbine technician) participates in the capacity of being the troubleshooter. The focus of the troubleshooting work is various qualities of the safety system, specifically the ones that exhibit anomalies. The potentiality of such anomalies is represented as vulnerabilities[5]. The troubleshooting process has as output recordings of the findings, and possible corrective work that may include part replacements. The recordings of the findings, actions and outcomes have an aboutness relation to all of the foregoing, as further covered in Section 4.5.

4.4 Sites and environments

BFO provides the entity *site* to represent three-dimensional immaterial entities that are determined relative to locations and boundaries of material entities. We define a *wind farm site* as a site that encompasses the wind farm and its immediate surroundings. In general, site boundaries need not be precisely defined, but for specific use cases they sometimes do.

The definition of the wind farm site allows us to introduce the *wind farm site environment* as the aggregate of material entities that naturally occupy the site (i.e. are not part of the wind turbines, balance of plant or site personnel). The site environment is not a proper universal in an ontological sense, since it does not have a strong identity and constitution (e.g. the portions of air that occupy the site are constantly changing), but it is an important defined class for the purpose of distinguishing between internal and external factors. For example, we want to be able to separate measurement signals that correspond to external conditions (e.g. wind speed) from those that correspond to internal system states.

4.5 Information content entities

BFO itself does not directly address information content in its scope, but provides the entity *generically dependent continuant* as a basis for building ontologies of information content entities. Such an entity can exist in multiple copies and does not depend on one specific bearer for its existence, but only depends on there being some bearer. Different mid-level ontologies have taken somewhat different approaches to the representation of information content, with the two most prominent being the Information Entity Ontology (IEO) within CCO and the Information Artifact Ontology (IAO) widely used within the Open Biological and Biomedical Ontology (OBO) Foundry ontologies. We have primarily made use of the former, as it is most applicable to our use cases.

Within an O&M context, the relevant information spans the diversity described in the introductory section. Some is created during the operational phase (e.g. event data) and some is carried from previous phases (e.g. commissioning reports). The OpOn framework strives to accommodate all relevant sources and types of information, with their various representations as data of different modes and structure. The explicit ontological modelling of information content brings the advantage of clearly separating material entities, their properties and the processes in which they participate from information about those entities.

[5] Vulnerabilities are negative or undesirable dispositions, as further discussed later in this paper.

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In contrast to traditional information systems, which do not aim for ontological reasoning capacity, the distinction between information and material entities is key for enhanced reasoning capabilities. The information content entities pertaining to O&M are introduced and defined in relation to physical aspects of the wind energy O&M domain via aboutness relations, along with other appropriate types of relations. Data can be annotated using entities and relations from the ontology, to tap into ontology-based reasoning, or simply to compare and align different datasets. The following sections provide concrete examples.

4.6 Alarms

Alarms are central to many wind turbine O&M activities, including our use case. This section presents an ontological framework for representing relevant entities, starting at the point of the information, and then moving on to define the entities that the different types of information are about.

Because the use case example in this paper considers only alarm logs and not broader event logs, and for the purpose of brevity, we focus in what follows exclusively on alarms. All the alarm-related definitions below can be understood as specialisations of more general event-related definitions.

We first distinguish the following information content entities relevant to alarms.

- An *alarm type description* is a *descriptive information content entity* that describes a specific kind of alarm that an alarm system is built to emanate under specific circumstances, and which may include information about underlying vulnerabilities that the alarm aims to detect, details about measurement equipment used for input data, details about the triggering logic and recommended response procedures. It is designated with an unambiguous alarm type identifier for referencing.
- An *alarm type identifier* is a *designative information content entity* that unambiguously designates a specific kind of alarm that an alarm system is built to emanate under specific circumstances. The alternative term *alarm code* may also be used.
- An *alarm status record* is a *descriptive information content entity* that describes the status of a particular kind of alarm in a specific alarm system at a point in time. It may describe activation, de-activation or the status at some point or period in time. It is often constructed as a combination of an alarm type identifier, a status identifier and timestamp(s). The alternative term *alarm log entry* may also be used in the context of an individual item within an alarm log.
- An *alarm log* is a *descriptive information content entity* that comprises a collection of alarm status records. It may or may not aggregate data from multiple alarm systems, and may or may not include filters to focus on specific alarm types, sub-systems, sub-components or time periods.

The term “alarm type” (or “kind of alarm”) broadly refers to a kind of alarm triggering situation. Such a type of situation may be rather general (e.g. over-speed alarm) or more specific (e.g. safety PLC input error). It may be generic and applicable to many different types of components (e.g. fan failure detected) or be defined with reference to a specific component (e.g. safety network hub offline). We do not introduce “alarm type” as an entity in the ontology, because it commonly refers to two different things that we want to distinguish between: types of alarm processes and types of alarm system capabilities[6]. As an example of the former sense of the term, all processes in which over-speed alarms have been emanated are instances of the universal *over-speed alarm emanation process.* In the latter sense, the alarm system bears a capability that is an instance of the universal *over-speed alarm emanation capability.* An alarm list provided by an OEM as part of the technical specifications for a wind turbine describes intended alarm system capabilities rather than any actual alarm processes. Many of the alarms that the system is capable of emanating may never be triggered in practice.

We introduce the following entities as relevant for capturing what alarm information is about.

- An *alarm system* is a *system* comprising engineered artifacts with the function to detect, and/or receive data about, alarming conditions and to emanate associated alarm status records and/or other alarm signals. It may have additional functions to store and process data, and to provide data services.
- An *alarm emanation capability* is a *capability* that inheres in an alarm system to detect and/or receive data about alarming conditions and to emanate associated alarm status records and/or other alarm signals.

[6]We adopt the ontological entity *capability* in the sense of a positive disposition, as proposed by Beverley et al. [13], noting that the alarm system operator has an “interest in” receiving alarm status information.

- An *alarm emanation process* is a *process* in which an alarm system participates and that is the realisation of some alarm emanation capability of that system.

An alarm type description and an alarm type identifier have an aboutness relation to an alarm emanation capability, but not directly to an alarm emanation process. An alarm status record has an aboutness relation to both a capability and a process.

Figure 3 and Figure 4 provide illustrations of the taxonomical structure and horizontal relations, respectively, for some entities relevant to the use case.

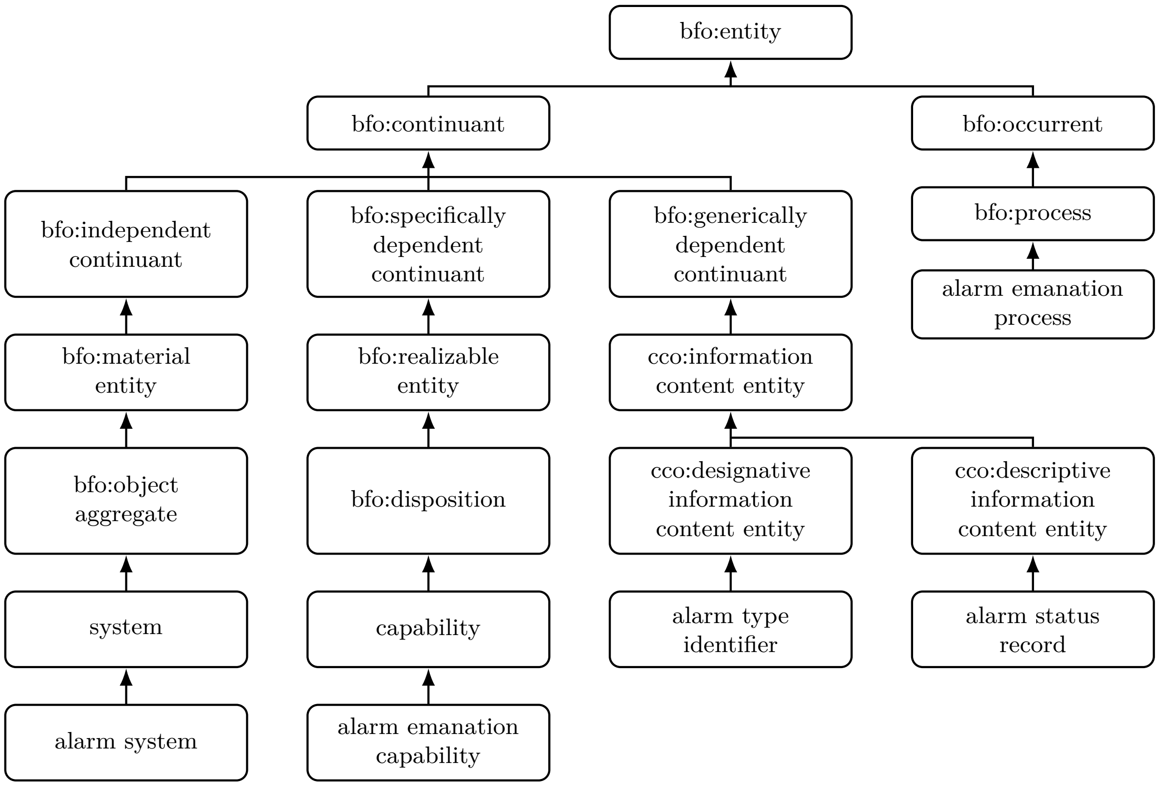

Figure 3: The taxonomical structure of an alarm-related slice of the ontology

4.7 Faults, failures and vulnerabilities

Alarm systems are deployed to notify operators – and possibly other affected parties – of undesirable conditions of the operated entity that require attention. In many cases, a protection system will already have executed an automated response (e.g. activated a brake procedure and brought the turbine to a stop), and the alarm is emanated to make the operator aware of the stop and the requirements to allow restart. Sometimes stop alarms correspond to unavoidable situations that do not require intervention (e.g. stops for structural protection during high wind gusts) and sometimes there are underlying faults that require remedies. For the ontology to support diagnostic and troubleshooting workflows in response to alarms, it needs to be able to distinguish between external environmental factors and faults, and capture the entities involved in the fault pathology.

We introduce the following entities in relation to faults.

- A *fault quality* is a *quality* that may impede the capabilities of its bearer. The alternative term *anomaly* may also be used.
- A *fault process* is a *process* that deviates in a negative way from how it was planned to unfold, whereby some intended capabilities were either not realised or lost.

There is often a relation between a fault quality and a fault process. For example, a loose cable connection may be a fault quality of a safety system communication network, resulting in a fault process

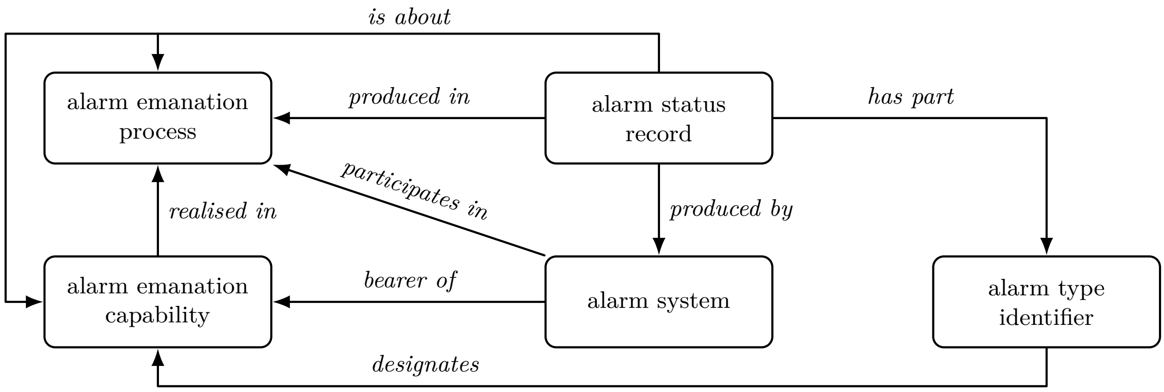

Figure 4: Relations between the entities of an alarm-related slice of the ontology

whereby network communication is intermittent. Some fault qualities may not immediately manifest as fault processes. For example, a faulty emergency brake system would only bring about a fault process when being deployed. Critical components often have a health monitoring system in place for trying to identify fault qualities before they cause fault processes, as is typically the case for a rotor over-speed protection system. Some fault processes may be difficult or impossible to tie to a specific set of fault qualities, due to a complex interaction of internal and external factors.

The term *failure* means the loss of function of a system or component. For example, when a safety system communication network has failed, it can no longer fulfil the function of providing communication between safety system components. Diagnostics and troubleshooting are often concerned with identifying and rectifying the fault quality that caused a failure.

Known potential fault qualities and processes are addressed at a protection and alarm system design stage. Such a potentiality is represented in the ontology as a *vulnerability*, which may at some point be realised, leading to fault qualities and/or processes, or may never be realised. A vulnerability is broadly equivalent to a failure mode. A vulnerability may be related to associated fault qualities and/or fault processes that result from or are symptomatic of its realisation. A fault quality or process may be linked to a work procedure, to prescribe a process to resolve it.

4.8 Relations for alarm annotations

To support useful reasoning about alarm log entries, we introduce the following relations.

- An *attributed to* relation to the detection or protection system where the alarm has its origin;
- An *applies to* relation to the material entity that is under monitoring and the alarm concerns;
- An *addresses* relation to the vulnerability that the alarm is intended to notify realisation of; and
- An *indicates* relation to the fault quality or process that the alarm suggests the presence of.

For example, a capability of an alarm system to emanate alarms in response to a detected rotor over-speed event is *attributed to* the rotor over-speed detection system, *applies to* the rotor system, *addresses* the vulnerability of the rotor to experience structurally dangerous levels of rotation, and *indicates* a rotor over-speed fault process. As part of the same sequence of events, the rotor over-speed protection system may trigger a related alarm, to record that this protection system has activated a brake system to bring the turbine to a stop. The capability of the alarm system to emanate alarms in response to rotor over-speed protection system brake activation is *attributed to* the rotor over-speed protection system. It has the same target for *applies to* and *addresses* as the first alarm emanation capability. It does not indicate one specific fault quality or process, but is rather an umbrella for a large number of such qualities and processes that are related to the state and health of the protection system.

There are many additional types of entities to which a relation can be captured, to further enrich alarm data processing. We focus here on a limited set of relations for the purpose of illustration.

Fundamental relations that apply in general across the domain can be captured in a reference ontology, whereas more specific details are captured in an application ontology within each organisation (e.g. concerning how a specific type of sensor is used in the rotor over-speed protection system or a particular turbine platform). Further to the annotation of the alarm list, individual entries in the alarm log can

On	Off	Identifier	Description
2025-04-20 19:01:15.907		2002	Safety PLC input error
2025-04-20 19:01:15.907		2001	Safety PLC communication error
2025-04-20 19:01:05.367		3004	Over-speed protection system alarm
...	(remote reset)	...	...
2025-04-20 18:54:35.637	2025-04-20 18:56:47.183	2000	Safety network hub offline
2025-04-20 18:54:35.637	2025-04-20 18:54:46.197	2001	Safety PLC communication error
2025-04-20 18:53:17.947	2025-04-20 18:53:18.933	2002	Safety PLC input error
2025-04-20 18:53:00.693	2025-04-20 19:01:20.923	3004	Over-speed protection system alarm

Table 2: Example excerpt from an alarm log focused on the over-speed protection system

be annotated with conclusions of root cause fault qualities and/or fault processes, to capture operational experience and make it easily accessible in future troubleshooting situations.

5 Use case demonstration: ontology-based diagnostics

We now explore how the ontology presented in Section 4 can be leveraged to automate the troubleshooting workflow described in Section 3.2 in a transparent and intuitive way. We describe how available data is annotated using entities and relations from the ontology, then demonstrate how a reasoner can traverse the knowledge base to isolate the most plausible fault explanation, and finally show how the ontology supports human-in-the-loop troubleshooting.

5.1 Data annotation

The first step is to annotate the available data using entities and relations from the ontology. This is an investment of effort that the user needs to make, but the process can be streamlined and potentially semi-automated. In return for this investment, the user has the data standardised and enriched with the context provided by the ontology and can leverage ontology-based reasoning. The tools for annotation must enable the user to capture knowledge about the alarms and associated entities in an intuitive and convenient way, using familiar terminology.

Table 2 provides an anonymised example excerpt from the alarm log, focused on the rotor over-speed protection system. It follows a common format, where the alarm start and end timestamps are included for each entry[7]. The equivalent portion of the alarm list contains each item without the on and off timestamps[8]. Each item in the alarm list is annotated as an instance of *alarm type description* and each item in the alarm log is annotated as an instance of *alarm status record*, establishing the link from a record in the alarm log to the corresponding alarm list entry[9]. Additional details such as the turbine platform, model and software version can be attached by the user to the alarm list as a whole to reflect organisational context.

Next, the annotations for the relations *attributed to*, *applies to*, *addresses*, and *indicates*, introduced in Section 4.8, are further applied. The three alarms[10] that refer to aspects of safety systems are *attributed to* the health monitoring of the safety system, *apply to* various components of the safety communication system, *address* different vulnerabilities of the safety system to communication faults, and *indicate* various fault qualities and fault processes of the safety network communication system. The health monitoring system is a protection system of the safety systems. If any critical safety system is not healthy, the turbine cannot be safely operated.

These annotations are based on the assumption that a knowledge base exists that captures key relations between turbine system components. For this use case, such relations include the fact that the rotor over-speed protection system is part of the overall turbine safety system, and shares the safety network with other components of that system.

[7]Another common format is for alarm start and end records to be separated.

[8]For the purpose of brevity, the alarm list is not repeated as a separate table.

[9]In practice, it is not necessary to manually annotate each individual item in a long list or document, but the annotation layer can be constructed flexibly.

[10]Strictly, it is the alarm emanation capabilities for which the relations are defined, but we relax the language for better readability, and a user-facing annotation tool can make similar simplifications, provided the meaning is unambiguous.

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5.2 Reasoning

Traditional process-oriented diagnostic and troubleshooting workflows such as the one illustrated in Figure 2 follow predetermined decision trees or run-books – a fixed sequence of checks, actions, and branches driven by specific triggers – requiring the user to know which procedure to follow and to supply technical detail at each step. Ontology-based approaches invert this relationship, offering a semantically-rich knowledge base which enables a reasoner to accept high-level queries and autonomously infer relevant context and data to generate plausible explanations from available evidence. Where process-oriented approaches are rigid and narrowly scoped to anticipated fault paths, ontology-grounded reasoning is exploratory and context-aware, capable of surfacing non-obvious cross-system correlations. Furthermore, ontologies can be used to annotate traditional documents to connect them to the relevant entities, as discussed in Section 5.3.

An ontology-based reasoner approaching the annotated data in Table 2 does not need to be provided with a specific question about rotor over-speed alarms, but can be provided with a general question about finding the cause of the active alarms, and will infer the details from the contextualised data. Using the *attributed to* relation, it groups these alarms by the originating system. Three of the four alarms are attributed to the safety system health monitor, while the fourth is attributed to the rotor over-speed protection system. The *addresses* relation reveals that the three safety system alarms all address vulnerabilities related to safety system communication faults, while the fourth addresses the broader vulnerability to rotor over-speed. From the functional dependencies captured in the broader knowledge base, the reasoner can infer that the rotor over-speed protection system depends on the safety system network, such that it is plausible that all four alarms are related. The absence of any other alarm related to the rotor over-speed protection system also suggests that it triggered an alarm because it lost communication rather than an actual over-speeding event or risk. The re-occurrence of the same alarms after a reset indicates that it is a permanent rather than transient issue.

The example here is based on a single data source, for the purpose of illustration. An ontology-based approach naturally supports integrating data from different sources within the same reasoning chain. For example, metric signals could be annotated in a similar way to alarms, and brought into the diagnostics. The integration of diverse O&M data enables a more comprehensive assessment of turbine health, laying semantically consistent foundations for further applications that support intelligent asset management.

5.3 Troubleshooting

When automated reasoning alone is insufficient – whether due to limited evidence or because resolution requires physical intervention such as resetting alarms, cycling components, or performing inspections – the ontology provides the semantic context needed to guide human-in-the-loop troubleshooting, ensuring technicians begin not from a generic procedure but from a contextually narrowed starting point informed by what has already been inferred and ruled out. Detailed technical specifications and process-oriented workflows that are available in traditional document form are linked to the appropriate context by annotations (e.g. linking technical specifications and fault investigation procedures for the safety PLC to the representation of that system component in the knowledge base, and to relevant vulnerabilities, fault qualities and fault processes).

5.4 Auditability and continuous improvement

The ontology allows for structured documentation of the entire process, including data collected, analysis performed, decisions made, and actions taken. This documentation can be valuable for future reference, training, and continuous improvement of maintenance procedures.

6 Towards ontology-grounded AI

The vision of the Operations Ontology proposed in this paper provides the initial foundations for more than just a shared vocabulary and data integration framework – it provides structured and machine-interpretable domain knowledge in a way that can play a pivotal role in shaping the future role of AI in O&M. In particular, such information can be used as context for AI models to help ensure their accuracy, transparency and scalability in the industry[11]. Below, we highlight the potential future directions for both academia and industry to build impactful AI applications on top of the OpOn.

[11] Besides advocates from within the specialist AI research community and among niche software providers, this ontology-powered approach has recently gained some traction in an enterprise software context, with products such as Microsoft Fabric IQ (https://blog.fabric.microsoft.com/en-us/blog/from-data-platform-to-intelligence-platform-introducing-microsoft-fabric-iq).

- Structured context for LLMs – The OpOn can be leveraged for grounding LLMs in a coherent domain-specific context (e.g. providing precise definitions for technical and commercial O&M terminology and relations between terms), improving accuracy, interpretability and resource-effectiveness, whilst limiting the chances for hallucinations. The ontology can also act as a source of truth, against which any LLM-generated content can be validated.
- Knowledge graph integration for deterministic reasoning – Knowledge graphs based on the ontology can be exposed to AI agents through interfaces such as Model Context Protocol (MCP), which allow the agents to focus on verified and curated data for deterministic inference. This would separate reliable symbolic reasoning from the flexible (but less predictable) outputs of the generative layer, which typically requires careful testing and monitoring. Note that the deterministic nature would be accomplished based on the symbolic layer (such as knowledge base queries, rules engines and constraint-based checks), wherein the outputs would be repeatable given the same set of inputs and the graph state.
- Hybrid neuro-symbolic reasoning in safety-critical applications – Hybrid approaches can be developed that combine formal reasoning approaches (e.g. rule-based and physics-based models) with trained models which are able to process unstructured data (e.g. maintenance reports and work orders). In comparison to purely statistically learned approaches, such formal reasoners are able to signal situations when a conclusion cannot be justified based on the available context, which can be highly valuable in situations that require high accuracy, such as critical engineering workflows.
- Probabilistic modelling of uncertain events – In O&M, there are countless scenarios that have an inherently probabilistic relationship, e.g. the chances for a fault type to progress to failure are dependent on the operational context and asset condition. The OpOn can help define the structure as well as scope of such relationships, with statistical models used in conjunction to estimate uncertainty, thus combining symbolic precision with statistical learning.
- Accelerated ontology and knowledge base development – AI agents can play an integral role in supporting domain-experts in proposing new ontology content (e.g. classes and axioms), mapping instance data to ontologically shared terms, and identifying inconsistencies or erroneous relationships. Note that while agents can help make the development process of knowledge assets more efficient, they should be implemented with a human-in-the-loop for validation during the build process.
- Data harmonisation for enabling model training – By identifying abstract commonalities between heterogeneous datasets (e.g. from different sites and operators, OEMs, asset types, etc.) based on shared foundational terms from the ontology, disparate sources can be merged into a larger and more consistent data corpus for training predictive models. This is especially valuable since large amounts of operational data lie in silos across organisations, with the sparsity limiting the performance as well as the generalisation abilities of machine learning (ML) models.
- Ontology-based validation and quality assurance for AI models – AI-generated artefacts (e.g. fault classification predictions and work order descriptions) can be validated by comparing against ontological constraints and logical axioms. This helps enable internal consistency as well as domain-specific validity of the outputs, providing an interpretable validation to complement learned behaviour of statistical models, and preventing unexpected errors from further propagation.

7 Discussion, conclusion and future outlook

We have presented the foundation for an effective ontology-driven approach to enhance the integration and understandability of data in wind energy O&M, with illustrations from a troubleshooting use case. The ontology-driven approach is expected to deliver the most significant advantages when:

- applied at a larger scale, connecting the full range of alarms and related troubleshooting procedures across a range of turbine platforms in a portfolio;
- adopted across stakeholders to align on controlled and shared vocabularies to facilitate information exchange; and
- built into information systems, so that they fully support standardised formats and contexts.

Through faster and more accurate reasoning over large and diverse datasets, the approach can enable better decision-making and attribution reporting, operational cost reductions and performance optimisations. With digitalisation and AI-readiness of increasing importance to the field, we propose OpOn as an

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initial step towards making semantically structured and consistent data a key foundation for intelligent O&M tools in both academia and industry.

As the OpOn continues to develop, we have identified potential areas for future development which include representations of:

- Further fault pathology;
- Wind farm and wind turbine sub-system and component breakdowns;
- Measurement equipment and systems relevant to wind energy O&M; and
- Computational models used in wind energy O&M and the results of such models.

As mentioned in Section 1, OpOn is designed to complement and be compatible with existing standards, such as the ISO/IEC 81346 RDS standards, not replace them. At the same time, OpOn is not tied to one particular standard or approach. This is in-keeping with the scope outlined in Section 2, ensuring that the ontology is both stable and flexible to different practices and industrial changes. We look forward to engaging with stakeholders to raise awareness and facilitate adoption, as industry engagement, broad collaboration and a long-term commitment are the key pillars for success.

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